

CLOUD COMPUTING

Unit I: Introduction to Cloud Computing

Cloud Service & Deployment Models



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Unit I – Topics Covered

- Introduction to Cloud Computing & Cloud Service Providers
- Definition and 5 Essential Characteristics of Cloud Computing
- Evolution of Computing – Parallel, Cluster, Distributed, Grid
- A Brief History of the Internet
- Cloud Architecture
- Cloud Service Models – IaaS, PaaS, SaaS
- Evolution of Cloud Computing
- Cloud Deployment Models – Public, Private, Hybrid, Community



What is Cloud Computing?

- On-demand delivery of computing resources – servers, storage, databases, networking, software – over the Internet
- Users access resources over the internet instead of owning and maintaining physical infrastructure
- Follows a pay-as-you-go pricing model, much like utilities such as electricity or water
- Resources can be provisioned and released rapidly, with minimal management effort
- Enables scalability, flexibility, and cost efficiency for organizations of any size



Examples of Cloud Computing

- **Storage & Backup:** Google Drive, Dropbox, Microsoft OneDrive
- **Email & Productivity:** Gmail, Microsoft 365, Google Workspace
- **Streaming Services:** Netflix, Spotify, YouTube
- **Social Media Platforms:** Facebook, Instagram, Twitter/X
- **Enterprise Applications:** Salesforce CRM, SAP on Cloud



Cloud Service Providers

- **Amazon Web Services (AWS):** market leader, offers the widest range of cloud services
- **Microsoft Azure:** strong enterprise integration and hybrid cloud support
- **Google Cloud Platform (GCP):** known for data analytics, AI and ML capabilities
- **IBM Cloud:** focused on enterprise and hybrid cloud solutions
- **Oracle Cloud & Alibaba Cloud:** growing global providers



Definition of Cloud Computing

As per NIST (National Institute of Standards and Technology)

- A model for enabling convenient, on-demand network access to a shared pool of configurable computing resources
- Resources – networks, servers, storage, applications, and services – can be rapidly provisioned and released
- Requires minimal management effort or service-provider interaction
- Widely accepted as the standard, formal definition of cloud computing
- Forms the basis of the 5 essential characteristics, 3 service models, and 4 deployment models



5 Essential Characteristics of Cloud Computing

- **On-Demand Self-Service:** users provision computing resources automatically, without human interaction
- **Broad Network Access:** services are available over the network through standard devices (laptops, mobiles, tablets)
- **Resource Pooling:** provider resources are pooled to serve multiple consumers using a multi-tenant model
- **Rapid Elasticity:** resources can scale up or down quickly to match demand
- **Measured Service:** resource usage is monitored, controlled, and billed on a pay-per-use basis



Evolution of Computing

Parallel Computing

Multiple processors execute tasks simultaneously within a single system to increase overall processing speed.

Cluster Computing

A group of interconnected, standalone computers work together and behave as a single, unified system.

Distributed Computing

Independent computers, located across a network, coordinate and communicate to solve a common problem.

Grid Computing

Geographically distributed computing resources are pooled together to solve large-scale computational problems.



A Brief History of the Internet

- **1960s:** ARPANET developed by the U.S. Department of Defense – the origin of packet-switched networking
- **1970s–80s:** The TCP/IP protocol suite is standardized, enabling networks to interconnect
- **1989–91:** Tim Berners-Lee invents the World Wide Web (WWW)
- **1990s:** Commercialization of the internet and the rise of web browsers
- **2000s onward:** Broadband and mobile internet accelerate global connectivity
- **Today:** The internet forms the foundational infrastructure that makes cloud computing possible



Cloud Architecture

Front End (Client Side)

The interfaces and applications end users interact with to access cloud services – web browsers, mobile apps, and thin or thick client software.

Back End (Provider Side)

Servers, storage, databases, virtual machines and applications managed by the provider. A network – usually the Internet – and middleware connect the front end to the back end.



Cloud Service Models

- **Infrastructure as a Service (IaaS):** provides virtualized computing resources – servers, storage, and networking – over the internet
- **Platform as a Service (PaaS):** provides a ready platform with tools and an environment for building, testing, and deploying applications
- **Software as a Service (SaaS):** delivers fully functional software applications over the internet on a subscription basis
- **Together:** these form the cloud computing stack, each offering a different balance of control versus convenience



Examples of IaaS, PaaS, SaaS

Infrastructure as a Service (IaaS)

- Amazon EC2
- Microsoft Azure Virtual Machines
- Google Compute Engine

Platform as a Service (PaaS)

- Google App Engine
- Microsoft Azure App Service
- Heroku

Software as a Service (SaaS)

- Gmail & Microsoft Office 365
- Salesforce
- Dropbox



Evolution of Cloud Computing

- **1960s:** John McCarthy proposes computing as a public utility
- **1999:** Salesforce launches enterprise applications via a website – an early form of SaaS
- **2002:** Amazon Web Services (AWS) is launched
- **2006:** Amazon EC2 and S3 bring IaaS to the mass market
- **2008:** Google App Engine and Microsoft Azure are launched, popularizing PaaS
- **Present:** Cloud computing is the backbone of modern IT, powering AI, big data, and mobile applications



Cloud Deployment Models

Public Cloud

Resources are owned and operated by third-party providers and shared among multiple organizations over the internet.

Private Cloud

Resources are dedicated to a single organization, offering greater control, customization, and security.

Hybrid Cloud

A combination of public and private clouds, allowing data and applications to move between them as needed.

Community Cloud

Shared infrastructure for a specific community of organizations with common concerns, such as security or compliance.



Unit I – Summary

- Cloud computing delivers on-demand IT resources over the internet with pay-as-you-go pricing
- Defined by 5 essential characteristics: on-demand self-service, broad network access, resource pooling, rapid elasticity, measured service
- Built on the evolution of parallel, cluster, distributed, and grid computing, enabled by the growth of the internet
- Offered through 3 service models – IaaS, PaaS, SaaS – and 4 deployment models – public, private, hybrid, community
- **Next:** Unit II – Virtualization Technology

